

Absolute Value

The absolute value of a number is its distance from zero, without considering its sign. This distance is always non negative, signifying the magnitude of a number.

Mathematically, the absolute value of a real number x is written as $|x|$. If $x \geq 0$, then $|x| = x$; if $x < 0$, then $|x| = -x$. For example, the absolute value of -5 is 5, while the absolute value of 7 is also 7. The absolute value concept is key in mathematics for comparing magnitudes, solving equations, measuring distances, and working with inequalities.

Example 1: Solve $|2x + 1| = 7$

To solve the equation $|2x + 1| = 7$, is necessary to consider two cases.

- Case 1: The inner expression is positive, so we can remove the absolute value:

$$2x + 1 = 7$$

Solving for x , we have:

$$2x = 6 \quad \Rightarrow \quad x = \frac{6}{2} = 3$$

- Case 2: The inner expression is negative, so we flip the sign and remove the absolute value:

$$2x + 1 = -7$$

Solving for x , we have:

$$2x = -8 \quad \Rightarrow \quad x = \frac{-8}{2} = -4$$

Therefore, the solutions to the given equation are $x = 3$ and $x = -4$.

Example 2: Solve $|x - 3| = 5$

To solve the equation $|x - 3| = 5$, it is necessary to consider two cases:

- Case 1: $x - 3 = 5$

$$x - 3 = 5 \Rightarrow x = 5 + 3 \Rightarrow x = 8$$

- Case 2: $x - 3 = -5$

$$x - 3 = -5 \Rightarrow x = -5 + 3 \Rightarrow x = -2$$

Therefore, the solutions are $x = 8$ and $x = -2$.

Example 3: Solve $|x - 4| < 3$

To solve the inequality $|x - 4| < 3$, consider that the absolute value inequality translates into a double inequality.

Rewrite the absolute value inequality into a double inequality:

$$-3 < x - 4 < 3$$

Now, solve for x :

$$-3 + 4 < x < 3 + 4$$

$$1 < x < 7$$

Therefore, the solution to the inequality $|x - 4| < 3$ is:

$$x \in]1, 7[$$

This means that all values of x between 1 and 7 (excluding 1 and 7) satisfy the inequality.

Example 4: Solve $|3x - 6| \geq 9$

To solve the inequality $|3x - 6| \geq 9$, we consider two cases:

- Case 1: $3x - 6 \geq 9$

$$3x \geq 15 \quad \Rightarrow \quad x \geq 5$$

- Case 2: $3x - 6 \leq -9$

$$3x \leq -3 \quad \Rightarrow \quad x \leq -1$$

Combining these cases, the solution to $|3x - 6| \geq 9$ is:

$$x \in (-\infty, -1] \cup [5, \infty)$$

This means that x must be less than or equal to -1 or greater than or equal to 5 .